



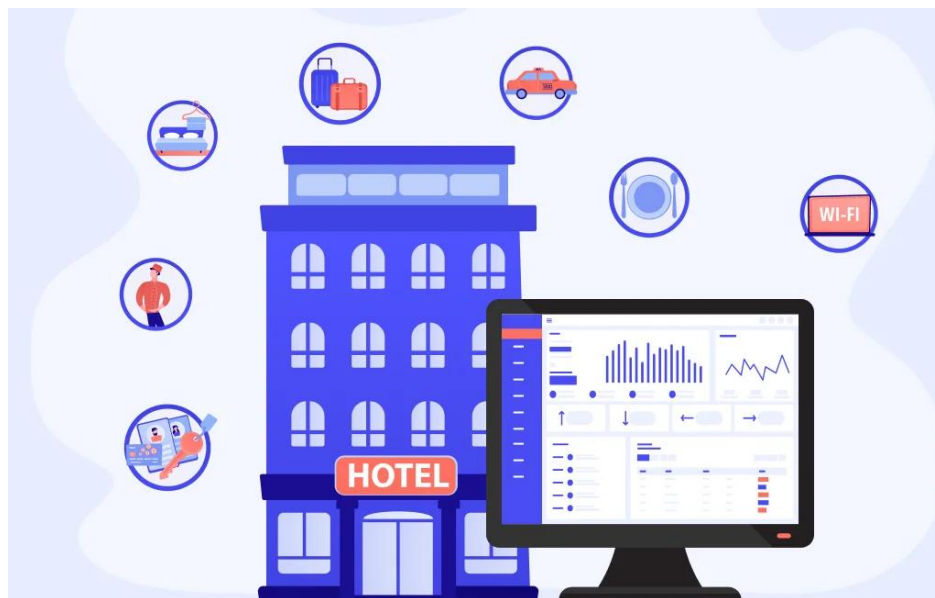
Department of Computer Science

Faculty of Applied Science

UNIVERSITY OF SRI JAYEWARDENAPURA

HOTEL MANAGEMENT SYSTEM

Part 02



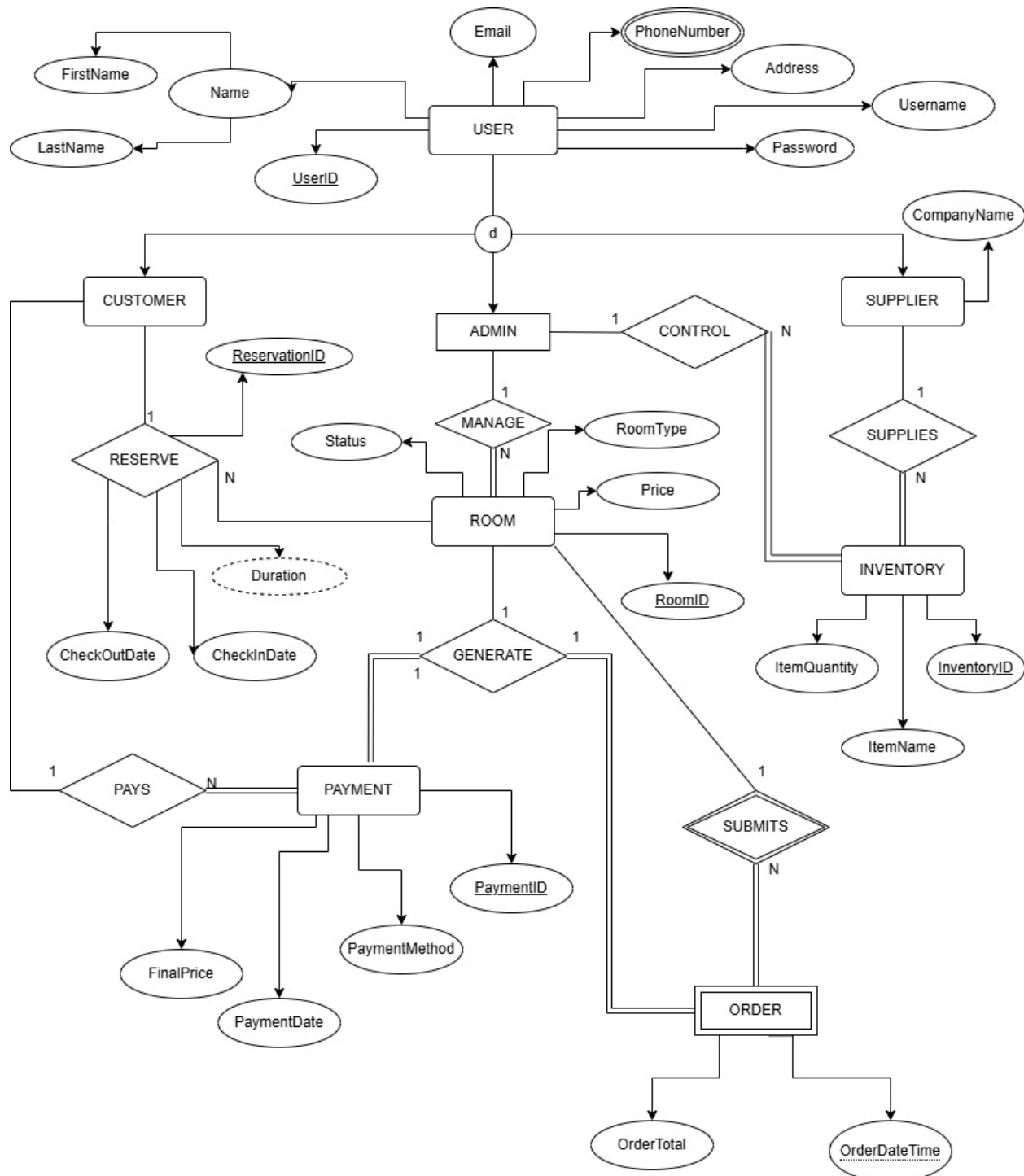
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- **Proposed real world problem**

The system efficiently controls and manages all hotel operations including rooms, reservations, food orders, payments, inventory, and suppliers. It follows an Enhanced Entity Relationship (EER) model in which USER acts as a supertype, and CUSTOMER, ADMIN, and SUPPLIER are defined as its subtypes, each having specific roles and responsibilities within the system.

- **EER Diagram**



• Entities and Attributes

USER (Strong entity)

- UserID (Primary Key)
- FirstName
- LastName } Name (Composite attribute)
- Username
- Email
- PhoneNumber (Multi value attribute)
- Address
- Password

CUSTOMER (Strong entity)

- UserID (Primary Key)

ADMIN (Strong entity)

- UserID (Primary Key)

SUPPLIER (Strong entity)

- UserID (Primary Key)
- CompanyName

ROOM (Strong entity)

- RoomID (Primary Key)
- RoomType
- Price
- Status

ORDER (Weak entity)

- OrderDate/Time (Partial key)
- OrderTotal

PAYMENT (Strong entity)

- PaymentID (PrimaryKey)
- PaymentDate
- PaymentMethod
- FinalPrice

INVENTORY (Strong entity)

- InventoryID (Primary Key)
- ItemName
- ItemQuantity

• Relationship Entities Cardinality Participation Attributes

Relationship	Entities	Cardinality	Participant	Attributes
RESERVE	CUSTOMER - ROOM	1: N	Customer → Partial Room → Partial	<u>ReservationID</u> CheckInDate CheckOutDate Duration
PAYS	CUSTOMER - PAYMENT	1: N	Customer → Patial Payment → Total	
GENERATE	ROOM – PAYMENT ORDER - PAYMENT	1: 1 1: 1	Room → Patial Order → Total Payment → Total	
SUBMIT	ROOM - ORDER	1: N	Room → Patial Order → Total	
SUPPLIES	SUPPLIER - INVENTORY	1: N	Supplier → Partial Inventory → Total	
MANAGE	ADMIN - ROOM	1: N	Admin → Partial Room → Total	
CONTROL	ADMIN - INVENTORY	1: N	Admin → Partial Inventory → Total	

• Constraints

In the system, every User must belong to exactly one of the three disjoint subclasses: Customer, Admin, or Supplier; no user can belong to more than one subclass and no other types of users are allowed.

- Each strong entity has a primary key.
- Each weak entity has partial key.
- Foreign keys connect:
 - ROOM → CUSTOMER
 - ORDER → ROOM
 - PAYMENT → CUSTOMER
 - PAYMENT → ROOM
 - PAYMENT → ORDER
 - INVENTORY → ADMIN
 - ROOM → ADMIN
 - INVENTORY → SUPPLIER
- A room cannot be reserved by more than one customer at the same time.
- Food orders can only be placed for occupied rooms.
- Payment must be made for each room stay.
- A room can be reserved or unreserved.

• Assumption

- Customers can order meals directly using their Room ID.
- The final payment is calculated by adding total room charges and total food order cost.
- Payment is calculated for each Room ID.
- One customer can reserve multiple rooms.
- A room can be reserved by only one customer at a given time.

- Relational Schema

